

Math 465: Introduction to Combinatorics

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Please fill out office hours survey by tomorrow (friday) night!

I will do my best to make slides available on canvas before class

Homework #1: due next Tuesday evening.

$\LaTeX$ file for homework is available in the “files” section on canvas

A note on participation

- Think out loud. Partial ideas, guesses, and incomplete arguments are all welcome.
- Be willing to be wrong. A big part of this class is figuring things out together, not just presenting finished answers.
- Give everyone time to think. If you know the answer right away, sometimes hold back for a moment. If you've already contributed a lot, let someone else jump in.
- Don't race ahead in the posted slides. They're there for accessibility and reference; please give yourself and your classmates the chance to solve things in real time.
- The goal is participation, not speed. I care much more about the ideas we develop together than about getting to the answer quickly.

Multinomial coefficients

Theorem

For an alphabet $\{x_1, \dots, x_j\}$ consisting of j symbols, there are

$$\binom{n}{n_1 \dots n_j} \stackrel{\text{def}}{=} \frac{n!}{n_1! \dots n_j!}$$

words of length n consisting of

n_1 copies of letter x_1 ,

.....,

n_j copies of letter x_j .

(Here $n = n_1 + \dots + n_j$.)

Proof

Use the Division Principle: permute n cards carrying those letters.

The numbers $\binom{n}{n_1 \dots n_j}$ are called *multinomial coefficients*.

A multinomial coefficient counts permutations of a *multiset*.

Anagrams

Problem

How many anagrams are there of the word STATISTICS?

Answer

$$\frac{10!}{1! \cdot 1! \cdot 2! \cdot 3! \cdot 3!} = 50400.$$

Problem

How many binary strings consist of three 0's and five 1's?

Answer

$$\frac{8!}{3! \cdot 5!} = 56.$$

Ordered set partitions

As before, assume that $n = n_1 + \cdots + n_j$.

Corollary

The number of ways to partition an n -element set S into disjoint subsets $S_1, \dots, S_j$ of sizes $n_1, \dots, n_j$, respectively, is $\binom{n}{n_1 \dots n_j}$.

Important: the subsets are numbered, i.e., labeled by $1, \dots, j$.
The elements within each subset are not ordered/labeled.

Example

The number of ways to assign n workers to j different tasks requiring $n_1, \dots, n_j$ workers, respectively, is $\binom{n}{n_1 \dots n_j}$.

Proof

Such ordered set partitions are encoded by permutations of multisets.

Definition

In the special case $j = 2$, we will use the simplified notation

$$\binom{n}{k} \stackrel{\text{def}}{=} \binom{n}{k, n-k} = \frac{n!}{k!(n-k)!}.$$

These numbers are called *binomial coefficients*.

Binomial coefficients play important roles in many applications.

One example is *binomial distributions* in probability theory.

Binary strings and subsets

Corollary

The number of binary strings of length n which consist of k entries equal to 1 and $n - k$ entries equal to 0 is $\binom{n}{k}$.

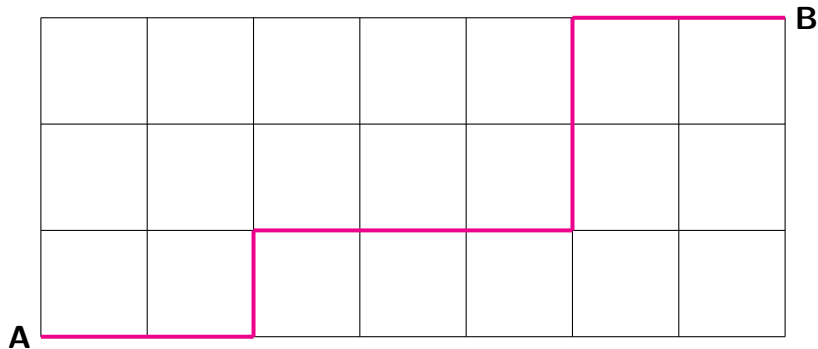
Theorem

The number of k -element subsets in an n -element set is $\binom{n}{k}$.

Example: $n = 4$, $k = 2$

0011	0101	0110	1001	1010	1100
$\{C, D\}$	$\{B, D\}$	$\{B, C\}$	$\{A, D\}$	$\{A, C\}$	$\{A, B\}$

Lattice paths



How many lattice paths are there from **A** to **B**?

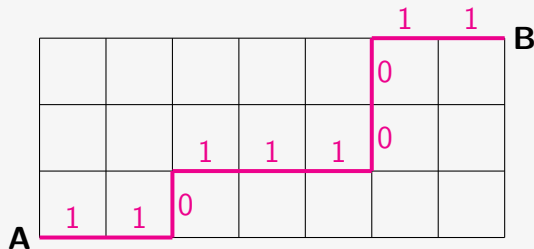
Lattice paths and binomial coefficients

Theorem

The number of lattice paths from $(0, 0)$ to (k, ℓ) is equal to $\binom{k+\ell}{k}$.

Proof

Such lattice paths are in bijection with the binary strings containing k entries equal to 1 and ℓ entries equal to 0 (or with k -element subsets of a $(k + \ell)$ -element set).



1 1 0 1 1 1 0 0 1 1
{1, 2, 4, 5, 6, 9, 10}

Counting lattice paths using the Addition Principle

1	4	10	20	35	56	84	120	B
1	3	6	10	15	21	28	36	
1	2	3	4	5	6	7	8	
1	1	1	1	1	1	1	1	

Pascal's recurrence

1	4	10	20	35	56	84	120 = $\binom{10}{7}$	B
1	3	6	10	15	21	28	36	
1	2	3	4	5	6	7	8	
1	1	1	1	1	1	1	1	
A	1	1	1	1	1	1	1	

Theorem

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

Alternative proofs of Pascal's recurrence

Theorem

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

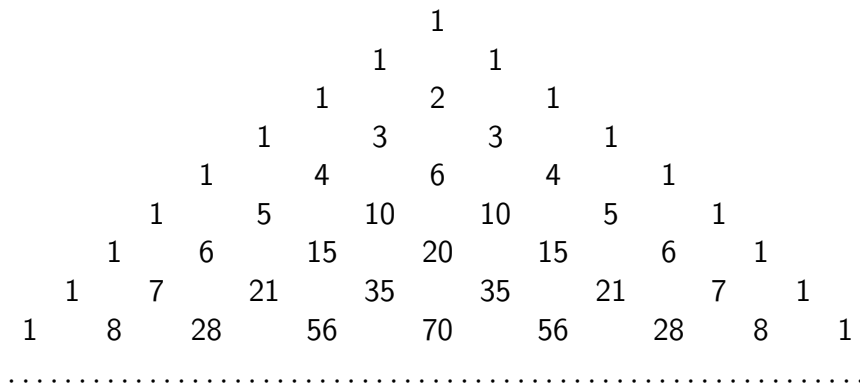
Proof #2

Fix an n -element set S . Select a distinguished element $s \in S$. Each k -element subset of S either contains s or it does not.

Proof #3

$$\begin{aligned}\binom{n-1}{k} + \binom{n-1}{k-1} &= \frac{(n-1)!}{k!(n-1-k)!} + \frac{(n-1)!}{(k-1)!(n-k)!} \\ &= \frac{(n-1)!(n-k+k)}{k!(n-k)!} = \binom{n}{k}\end{aligned}$$

Pascal's Triangle



[Blaise Pascal](#) (France, 1653) was the first to systematically describe various properties of this triangle (and apply it to probability theory).

The triangle was already known in Asia more than 1,000 years ago, thanks to [Halayudha](#) (India), [Al-Karaji](#) (Iran), and [Jia Xian](#) (China).

Compositions

Definition

A *composition* (of an integer n , with k parts) is a positive integer solution of the equation

$$x_1 + \cdots + x_k = n. \quad (*)$$

A *weak composition* is a nonnegative integer solution of $(*)$.

Example: $n = 5, k = 3$

Compositions:

(3, 1, 1) (1, 3, 1) (1, 1, 3) (2, 2, 1) (2, 1, 2) (1, 2, 2)

Weak compositions:

(5, 0, 0) (4, 1, 0) (1, 4, 0) (3, 2, 0) (2, 3, 0) (3, 1, 1) (2, 2, 1)
(0, 5, 0) (4, 0, 1) (1, 0, 4) (3, 0, 2) (2, 0, 3) (1, 3, 1) (2, 1, 2)
(0, 0, 5) (0, 4, 1) (0, 1, 4) (0, 3, 2) (0, 2, 3) (1, 1, 3) (1, 2, 2)

Counting weak compositions

Theorem

The number of weak compositions of n with k parts is equal to

$$\binom{n+k-1}{k-1}.$$

Example: $n = 2, k = 3$

$(2, 0, 0)$	$(0, 2, 0)$	$(0, 0, 2)$	$(1, 1, 0)$	$(1, 0, 1)$	$(0, 1, 1)$
○ ○	○ ○	○ ○	○ ○	○ ○	○ ○
0011	1001	1100	0101	0110	1010

Counting compositions

Theorem

The number of compositions of n with k parts is equal to $\binom{n-1}{k-1}$.

Proof

$$\left\{ \begin{array}{l} \text{compositions of } n \\ \text{with } k \text{ parts} \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \text{weak compositions of } n-k \\ \text{with } k \text{ parts} \end{array} \right\}$$

Example: $n = 5, k = 3$

$$\begin{array}{cccccc} (3, 1, 1) & (1, 3, 1) & (1, 1, 3) & (2, 2, 1) & (2, 1, 2) & (1, 2, 2) \\ (2, 0, 0) & (0, 2, 0) & (0, 0, 2) & (1, 1, 0) & (1, 0, 1) & (0, 1, 1) \end{array}$$

The number of these weak compositions is $\binom{(n-k)+k-1}{k-1} = \binom{n-1}{k-1}$.

Binomial identities (1)

There are many algebraic identities involving binomial coefficients. Behind each of them, there is a combinatorial correspondence.

To prove an identity, we either (a) establish a bijection between two sets, then enumerate each of them, or (b) “reverse engineer” an enumerative problem that can be solved in two different ways.

Proposition

$$\binom{n}{k} = \binom{n}{n-k}$$

Proof

There is a bijection between k -element subsets of an n -element set and $(n-k)$ -element subsets of the same set.